1, Rings and ideals

Ring: $(R, +, \cdot)$
- $(R, +)$ abelian group
- $(R, \cdot)$ semigroup
- distributive law

All our rings will have a unit $1 \in R$
All our rings unless specified will be commutative

$0 \in R$ neutral element under $+$
The trivial ring $\{0\}$ has $1 = 0$
$u \in R$ is a unit (= invertible element) if there exist $\exists v \in R.\ uv = vu = 1$. This v is unique $v = u^{-1}$

$R^\times$ or R^{-1} or $GL(R)$... units of R

Ring homomorphism is a map $\varphi: R \to R'$ that preserves sums, product and ones

...

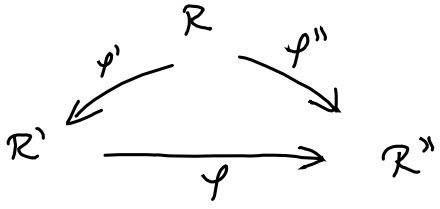
R-algebras: ring R' with a homomorphism

$\varphi : R \to R'$

called structure homomorphism

Every ring is a $\mathbb{Z}$-algebra

R-algebra homomorphism between R', R'' is a ring homomorphism that is compatible with structure maps



Examples:

- $\mathbb{Z}_2$
- R ring, Y set

  $R^Y = \{ Y \to R \}$ is a ring with point wise operations $f, g \in R^Y$ $(f+g)(y) = f(y) + g(y)$

  $(f \cdot g)(y) = f(y) \cdot g(y)$

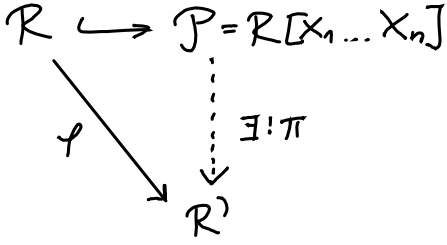
  identity: $y \mapsto 1$

Ring of polinomials: $\mathcal{P} = R[X_1, \dots X_n]$

$\mathcal{P}$ has the following universal property (UL):

$\forall$ hamamorphism $\varphi : R \to R'$. $\forall a_1, \dots a_n \in R$

$\exists!$ hamamorphism $\pi : \mathcal{P} \to R'$ s: $\pi_{|R} = \varphi \wedge \pi(X_i) = a_i$



$$\pi\left(\sum_i r_i X_1 \cdots X_n\right) = \sum_i \varphi(r_i) a_1 \cdots a_n$$

• Ideal: $I \trianglelefteq R$ $\quad 0 \in I$

$x, y \in I \Rightarrow x+y \in I$

$r \in R, x \in I \Rightarrow rx \in I$

$S \subseteq R$ $\quad \langle S \rangle$ ... Ideal of R generated by S

$\langle S \rangle = \{ \sum_{s \in S}^{\text{finite}} a_s s \; ; \; a_s \in R \}$

$\langle a \rangle$ ... principal ideals

Operations on ideals

• I_λ $(\lambda \in \Lambda)$; ideals of R

$$\sum_{\lambda \in \Lambda} I_\lambda = \{ \sum_{\lambda \in \Lambda}^{\text{finite}} b_\lambda \; ; \; b_\lambda \in I_\lambda \}$$

• $\cap I_\lambda \trianglelefteq R$ $\qquad \bigcap_{\emptyset} I_\lambda = R$

• $\prod I_\lambda = \langle \{ \prod_{\lambda \in \Lambda'} b_\lambda \; ; \; \Lambda' \overset{\text{finite}}{\subseteq} \Lambda, \; b_\lambda \in I_\lambda \} \rangle$

• $(I : J) = \{ x \in R \; ; \; xJ \subseteq I \}$ transporter

• $I \cdot J \subseteq I \cap J \subseteq I, J \subseteq I + J$

$I \subseteq (I : J)$

$I(J+K) = I \cdot J + IK$ DN

$I \trianglelefteq R$ is proper if $I \subsetneq R$

$I \trianglelefteq R$ is trivial if $I = \{0\}$

$\langle x \rangle = R \Leftrightarrow x \in R^\times$

($\Leftarrow$) obvs

($\Rightarrow$) $\langle x \rangle = R \Rightarrow \exists u \in R \quad ux = 1 \Rightarrow x \in R^\times$

$\varphi : R \to R' \qquad I \lhd R$

$\varphi(I)$ tipically not ideal (

$\mathbb{Z} \hookrightarrow \mathbb{Q}$

$\uparrow \quad \nearrow$

$2\mathbb{Z}$

$\langle \varphi(I) \rangle =: I^e$ "estention of I"

$I' \lhd R'$

$\varphi^*(I') \lhd R$ "kontraction of I'" $I'^c = \varphi^*(I')$

$\varphi : R \to R' \Rightarrow \ker \varphi = \varphi^*(0) \lhd R$

$\ker \varphi = \{0\} \Leftrightarrow \varphi$ is injective

$I \trianglelefteq R$ Form cosets $\{x+I ; x \in R\} = R/I$

$$(x+I)+(y+I) = (x+y)+I)$$
$$(x+I)(y+I) = xy+I$$

Quotient or factor ring

$$K : R \longrightarrow R/I \qquad x \mapsto x+I$$

canonical quotient homomorphism

K is surjective

$\ker K = I$

$$\boxed{R/\ker\varphi \cong \operatorname{im}\varphi}$$

$R \quad I \trianglelefteq R \quad I \subseteq J \trianglelefteq R \quad K : R \to R/I$

$$J/I = \{y+I \mid y \in J\} = K(J) \trianglelefteq R/I$$

maps $J \mapsto J/I$ in $J' \mapsto K^{-1}(J)$ are inverse hence bijection

$$\frac{R/I}{J/I} \cong R/J$$

Proposition: $\mathcal{P} = R[X]$; $a \in R$. Let $\pi: \mathcal{P} \to R$ be the R-ring homomorphism $X \mapsto a$

a) $\ker \pi = \{F \in \mathcal{P};\ F(a) = 0\} = \langle x - a \rangle$

b) $\mathcal{P}/\ker \pi \cong R$

Proof: $G := X - a$

$\forall F \in \mathcal{P}.\ \exists H \in \mathcal{P}.\ \exists r \in R\ \ni: F = GH + r$

B; linearity it suficies to prove this for $f = X^n$

$f = X^{n-1}X = X^{n-1}(G + a) = GX^{n-1} + aX^{n-1}$

Repeat this starting with X^{n-1}

a) $\pi(F) = \underset{0}{\pi(G)}\,\pi(H) + \underset{r}{\pi(r)} = r$

Hence $F \in \ker(F) \Leftrightarrow r = 0 \Leftrightarrow F(X-a) \cdot H$

$\Leftrightarrow F \in \langle X - a \rangle$

b) $\mathcal{P}/\ker \pi \cong \operatorname{im} \pi = R$

■

$\deg F = \max \{\deg(M);\ M \text{ monomial in } F\}$

$M = X_1^{\alpha_1} \dots X_n^{\alpha_n}$

$\deg(M) = \sum \alpha_i$

$\deg(0) = -\infty$

$\deg(F \cdot G) \leq \deg F + \deg G$

Idempotents if $e^2 = e$

Poglej si pri Filipu, če je potrebno

Prop: Let R be a ring and e', e'' complementary idempotents ($e''=1-e'$). Set $R'=Re'$ in $R''=Re''$

Define $\varphi: R \longrightarrow R' \times R''$

$$x \longmapsto (xe', xe'')$$

Then R is isomorphiseen $R' \cong R/Re''$, $R'' \cong R/Re'$

Proof:

Consider $\varphi': R \longrightarrow R$

$$x \longmapsto xe'$$

φ' is homomorphism $\varphi'(xy) = xye' = xye'^2 = \varphi(x)\varphi(y)$

$\ker \varphi' = R''$

$(\subseteq)$ $\varphi'(x)=0 \Rightarrow xe'=0$

$$x \cdot 1 = x(e'+e'') = 0 + xe'' \in R''$$

$(\supseteq)$ $\varphi(xe'') = x\underbrace{e''e'}_{=0} = 0$

Then $\operatorname{im}\varphi = R'$ and $R' \cong R/Re''$

Similarly we have $\varphi'': x \longmapsto xe''$

Then $\varphi = (\varphi', \varphi'')$ is homomorphism

$\ker \varphi = 0$

$x \in \ker\varphi \Rightarrow xe'=0$ and $xe''=0$

$\Rightarrow 0 = x(e'+e'') = x \cdot 1 = x$ ✓

$\operatorname{im}\varphi = R' \times R''$

$(xe', x''e'') = \varphi(x'e' + x''e'')$

$\Rightarrow \varphi$ is isomorphism

■

II Prime ideals

$x \in R - \{0\}$ is a zero devisor $\Leftrightarrow \exists y \in R . xy = 0$

$zdiv R = \{ x \in R \text{ zero devisor} \}$

Def $S \subseteq R$ is multiplicative if $1 \in S$ & $S \cdot S \subseteq S$

$P \trianglelefteq R$ is prime if $R - P$ is multiplicative

Eg: $P \trianglelefteq R$ is prime

Ring is a field if $0 \neq 1$ and an every non zero element is invertable

Ring where $\langle 0 \rangle$ is prime ideal is integral domain (od. helobar)

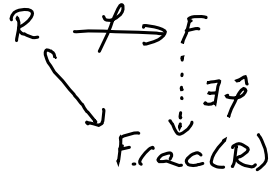
such ring has no zero devisors

- R integral domain $\Rightarrow$ can form a field of fractions

$$Frac(R) = \left\{ \frac{a}{b} ; a \in R ; b \in R - \{0\} \right\}$$

- every integral domain is a subring of a field
- every subring of a field is integral

UL for Frac(R): Each embeding of R into F Field extends to Frac(R) $\xrightarrow{\hat{\varphi}}$ F



$\hat{\varphi}\left(\frac{a}{b}\right) = \frac{\varphi(a)}{\varphi(b)}$

- Polynomial rings over integral domains

  R domain, $P = R[x]$

  P is also a domain and $\deg(FG) = \deg F + \deg G$

  We order P with so-called graded lexicographic (grlex) ordering:

  order first by degree, then lexicografically

  $x_1^4 x_2 < x_1^3 x_2^3 x_3 < x_1^2 x_2 x_3^4$

  Among all the monomials pick the largest one. This is LM(F) = leading monomial of F

  LT(F) = leading term of F $a \cdot LM(F)$ if $a \cdot LM(F)$ appears in F

  LC(F) = leading term of F a

$F, G \in P \Rightarrow LT(FG) = LT(F) \cdot LT(G)$

$LT(F) = a \cdot m$, $LT(G) = a \cdot n$

Then $a \cdot b \neq 0$ bcos R is a domain

$\deg(m \cdot n) = \deg(m) + \deg(n) = \deg(F) + \deg(G)$

if $m' \neq m$ or $n' \neq n$ then $m'n' <_{grlex} mx$ by max of m, n !

Hence $LT(F \cdot G) = abmn = LT(F) \cdot LT(G)$

P domain $\Rightarrow$ hen form $Frac(P) =: K(X)$,

$K = Frac(R)$ ↑ field of rational functions

$P^\times = R^\times$

if $F \cdot G = 1$ then $\deg(FG) = 0$

$\deg(F) + \deg(G) \geq 0$

lema: let $\varphi: R \to R'$ is homomorphism and $T \subseteq R'$.

a) if T is multiplicative then $\varphi(T)$ is too

b) if φ is surjective (onto), then also the converse holds

Proof:

a) $S = \varphi^{*}(T)$

Since $\varphi(1)=1 \in T$ we have $1 \in S$

For $x,y \in S$ we have

$$\varphi(xy) \in \varphi(S) \subseteq T$$
$$\| $$
$$\varphi(x)\cdot\varphi(y) \in \varphi(S)\cdot\varphi(S) = T\cdot T$$
$$\Rightarrow T\cdot T \subseteq T$$

b) Now assume $S = \varphi^{*}(T)$ is multiplicative

T is multiplicative

- $1 \in T$ since $1 \in S \wedge \varphi(1)=1$
- given $x', y' \in T$ then $\exists x, y \in S$ $x' = \varphi(x)$, $y' = \varphi(y)$
  Then $x\cdot y \in S$ and $\varphi(x,y) = \varphi(x)\,\varphi(y) = x'y' \in T$

Corollary:

$\varphi: R \to R'$ hom, $J \trianglelefteq R'$. $I := \varphi^*(J)$

a) if J is a prime ideal then so is I

b) if φ is onto (surj) $\wedge$ I is prime $\Rightarrow$ J is prime

Proof: Apply the lemma to the complement of J

Corollary: Let $P \trianglelefteq R$. Then

P prime $\Leftrightarrow$ R/P is a domain

Definition: $M \trianglelefteq R$ is a maximal ideal of $M \subsetneq R$ and $\nexists I \trianglelefteq R$. $M \subsetneq I \subsetneq R$

Proposition:

Ring R is a field $\Leftrightarrow$ (0) is a max ideal

Proof:

($\Rightarrow$) $\exists\, 0 \neq x \in I$. Then $\exists x^{-1} \in R$ Hence $1 = x \cdot x^{-1} \in I$

So $I = R$

($\Leftarrow$)

Pick $x \in R - (0)$ then $(0) \subsetneq \langle x \rangle \lhd R$

Since (0) is a max ideal $\langle x \rangle = R$

in particular $1 \in \langle x \rangle$ so $\exists y \in R.\ xy = 1$

So x is invertable

Coro. Let R be a ring. $M \lhd R$. Then M is max ideal $\Leftrightarrow$ R/M is a field

Proof: M is a max ideal $R \Leftrightarrow (0)$ is a max ideal R/M $\overset{\text{prop}}{\Leftrightarrow}$ R/M is a field ∎

Cor. Every max ideal is prime

Proof Every field is an int domain

Examples:

a) R int domain; $R[x,y] =: \mathcal{P}$. Then $\langle x \rangle \triangleleft \mathcal{P}$ is a prime ideal that is not maximal

$$\langle x \rangle \subsetneq \langle x, y \rangle \subsetneq \mathcal{P}$$

$\langle x \rangle$ is prime since $\mathcal{P}/\langle x \rangle = R[y]$ is a domain

$\langle x, y \rangle$ is max $\Leftrightarrow$ $\mathcal{P}/\langle x, y \rangle \cong$ is a field

b) R ring, $\mathcal{P} = R[x_\lambda ; \lambda \in \Lambda]$. For each $\lambda \in \Lambda$

$$M := \langle x_\lambda - a_\lambda ; \lambda \in \Lambda \rangle$$

Then $\mathcal{P}/M \cong R$

Theorem: Every proper ideal is contained in a maximum ideal

Proof:

$I \subsetneq R$

$\mathcal{S} = \{ J \subsetneq R ; I \subseteq J \}$

We want to use Zorn lema

- $I \in \mathcal{S} \Rightarrow \mathcal{S} \neq \emptyset$
- every chain has an upper bound

  chain $\{I_\lambda ; \lambda \in \Lambda\}$ in $\mathcal{S}$

  upper bound $J = \bigcup_{\lambda \in \Lambda} I_\lambda \subsetneq R$

  Since $1 \notin I_\lambda \ \forall \lambda \Rightarrow 1 \notin J \Rightarrow J \neq R$

According to zorn lema the set $\mathcal{S}$ has the max element (maximal ideal) and it containes I

Cor:

R ring, $x \in R$ Then

$x \in R^\times \Leftrightarrow x$ does not lie in max ideal

Proof

$x \in R^\times \Leftrightarrow \langle x \rangle = R \quad \Rightarrow$

$x \in R \Leftrightarrow \langle x \rangle \subsetneq R \Leftrightarrow \exists M \overset{\max}{\subset} R \; x \in R \subseteq M \subsetneq R$

II.1 Factorisation

R ring, integral domain

u unit $\Leftrightarrow \exists u^{-1} \in R \Leftrightarrow \langle u \rangle = R$

$a|b \Leftrightarrow \exists g \in R.\ b = g \cdot a \Leftrightarrow b \in \langle a \rangle \Leftrightarrow \langle b \rangle \subseteq \langle a \rangle$

a and b are associates $a|b \wedge b|a \Leftrightarrow a = bu \Leftrightarrow \langle a \rangle = \langle b \rangle$

a is a proper devision of $b \Leftrightarrow b = ag$, $a, g \notin R^\times$

$\Leftrightarrow \langle b \rangle \subsetneq \langle a \rangle \subsetneq R$

a is irreducable $\Leftrightarrow a \notin R^\times$ and has no proper devisors

$\Leftrightarrow$ maximal among principa ideals

p is prime element $\Leftrightarrow p \in R^* \wedge (p|ab \Rightarrow p|a \vee p|b)$

$\Leftrightarrow \langle p \rangle$ is prime ideal

Definition Domain R is UFD (unique factorisation domain) if

1) each element is a product of irreducables and units
2) factorisation is unique up to units and order

Prop: An int-domain R is UFD $\Leftrightarrow$

1) every ascending chain of principal ideals stabilazes
2) every ireducable element is prime

Proof:

$\Rightarrow$. if $\langle a_1 \rangle \subset \langle a_2 \rangle \subset \dots$

then $a_{i+1} | a_i \wedge a_i / a_{i+1}$ is not a unit

So the numbers of factors strictly decreses

This canot happe so 1 holds

- Suppose p is ireducable and $p | ab$ Fedor a, b. The uniqueness of factorisation

Filipovi zapiski

Def: A domain R is principal ideal domain (PID) if every ideal is principal

Prop: R PID. Then

a) Every irreducible element is prime

b) Every nonzero ideal is max

Proof:

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III Radical ideals

Def: Jacobson radical of ring R $\operatorname{rad}(R)$ is the intersection of all max ideals of R

$$\operatorname{rad} R = \bigcap_{M \trianglelefteq R \text{ max}} M$$

Prop: R ring $I \trianglelefteq R$ $u \in R^\times$. Then

$$\forall x \in R. \left(x \in \operatorname{rad} R \iff \forall y \in R.\ u - xy \in R^\times \right)$$

In particular

- $R^\times + \operatorname{rad} R \subseteq R$
- $1 - I \subseteq R^\times \Rightarrow I \subseteq \operatorname{rad} R$

Proof: Filip

Cor: $I \trianglelefteq R$, $K: R \to R/I$ quotient homomorphism.
Assume $I \subseteq \operatorname{rad} R$. Then

$$K|_{\operatorname{Idem}(R)} : \operatorname{Idem}(R) \longrightarrow \operatorname{Idem} R/I \quad \text{is injective}$$

Proof:

Let $e, e' \in \operatorname{Idem}(R)$, $K(e) = K(e')$

$x := e - e'$ then $x^3 = e^3 - 3e^2e' + 3ee'^2 - e'^3 =$

$= e - 3ee' + 3ee' - e' = e - e' = x$

$x - x^2 = 0$

$x(1-x) = 0$ we the have $K(x) = 0 \Rightarrow x \in I$

$\Rightarrow x \in \operatorname{rad} R$

By the proposition $1 - x^2$ is invertible, because

$x \in \operatorname{rad} R \Rightarrow x = 0$ $(x(1-x^2) = 0 \Rightarrow x = 0)$

Def: Ring is **local** if it has only one maximal ideal

R is called **semi-local** if it has finitely maximal ideals. The **residue field** of a local field (R, M) is R/M

(M: max ideal)

Lema: R ring, $N = R - R^\times$ the set of non units. Then

R local $\Leftrightarrow$ $N \lhd R$

(because N contains all non units)

Proof:

($\Leftarrow$) $I \subsetneq R$ is contained in N, because I contains only non units. if $N \lhd R$ then this must be a maximal.

($\Rightarrow$) R is local with M maximal, then

$R - M = R^\times = R - N \Rightarrow M = N \Rightarrow N \lhd R$

∎

Example

$R := R' \times R''$ is not local, because $(1,1) = (1,0) + (0,1)$

Example: R ring, $n \in \mathbb{N}$. Formal power series (FPS)

In $X_1 \dots X_n$ are formal (oo) series of the form

$$\sum_{i \in \mathbb{N}^1} a_i X_1^{i_1} \cdot \dots \cdot X_n^{i_n} \qquad a_0 := \text{const. coef.}$$

We add and multiply the same as polinomials

$R[[X_1, \dots X_n]]$

$I := \langle X_1 \dots X_n \rangle \triangleleft R[[X]]$

Then $\sum a_i X_1^{i_1} \cdot \dots \cdot X_n^{i_n} \longmapsto a_0 \in R$ ($\in R[[X]]$)

is a surjective homomorphism with kernel I

$R[X]/I \cong R$

$M \triangleleft R$ an ideal, $N := I + MR[[X]] \triangleleft R[[X]]$

$\Rightarrow R[[X]]/N \cong R/M$

Filip

$R[[X]]/N \cong R/M$

with isomorphism

$$\frac{R[[X]]}{I + MR[[X]]} \longrightarrow R/M$$

$$\sum a_i X^i \longmapsto$$

Prop: R ring. $S \subseteq R$ multiplicative set, $I \lhd R$
$I \cap S = \emptyset$.

Form $\mathcal{J} = \{ J \lhd R ; I \subseteq J \wedge J \cap S = \emptyset \}$

Then $\mathcal{J}$ contains maximal element and each such maximal element is a prime ideal

Proof: 1) $I \in \mathcal{J} \Rightarrow \mathcal{J} \neq \emptyset$

2) Let J_λ be a chain in $\mathcal{J}$. Define

$J = \bigcup J_\lambda \Rightarrow J$ is an ideal (because increasing chain of ideals is ideal)

Then $J \in \mathcal{J}$

$\Rightarrow J$ is an upper bound in $\mathcal{J}$

We use Zorn's lemma $\Rightarrow \mathcal{J}$ has maximal elements

Let $P \in \mathcal{J}$ be maximal element

P is prime.

Take $x, y \in R - P$. Then $P + \langle x \rangle$ and $P + \langle y \rangle$ aren't in $\mathcal{J}$ by maximality of P

This means $P + \langle x \rangle \cap S \neq \emptyset$

$\exists, p, q \in P. \exists a, b \in R.$ $p + ax \in S$ and $q + by \in S$

$\overset{S}{\underset{\text{multipl.}}{\Longrightarrow}}$ $(p+ax)(q+by) = \underbrace{pq}_{\in P} + \underbrace{pby}_{\in P} + \underbrace{qax}_{\in P} + \underbrace{axby}_{\in P} \in S$

$abxy \notin P$ because $P \cap S = \emptyset$

$\Rightarrow xy \notin P$

$\Rightarrow xy \in R - P$ ∎

Radical of ideals $I \subseteq R$

Its radical is $\sqrt{I} = \{x \in R ; \exists n \in \mathbb{N} ; x^n \in I\}$

- obviously $I \subseteq \sqrt{I}$, $\sqrt{\sqrt{I}} = \sqrt{I}$ and $J \subseteq I \Rightarrow \sqrt{J} \subseteq \sqrt{I}$

Def: $I \trianglelefteq R$ is radical $:\Leftrightarrow I = \sqrt{I}$

Lema:
if $I = \bigcap_\lambda P_\lambda$ P_λ prime, then I is radical.

Proof: Suppose $x \in \sqrt{I}$. Then $\exists n \in \mathbb{N}$
$x^n \in P_\lambda$ for all $\lambda \Rightarrow x \in P_\lambda \vee x \in P_\lambda \vee \ldots \Rightarrow x \in P_\lambda$
So $x \in \bigcap P_\lambda = I \Rightarrow x \in I$

Def: $nil(R) = \sqrt{\langle 0 \rangle}$; is the nil radical.
$I \trianglelefteq R$ is nilpotent $\Leftrightarrow \exists n \in \mathbb{N} . I^n = \langle 0 \rangle$

Since $radR = \bigcap_{M \trianglelefteq R} M$ is an intersection of prime ideals

- $radR$ is an radical ideal $\sqrt{radR} = radR$
- $\langle 0 \rangle \subseteq radR \Rightarrow \sqrt{\langle 0 \rangle} \subseteq \sqrt{radR}$

Def. ring R is reduced : if $nilR = \langle 0 \rangle$

Theorem: if $I \lhd R$, then $\sqrt{I} = \bigcap_{I \subseteq P \text{ prime}} P$

Proof. • let's say $x \notin \sqrt{I}$. Define $S = \{1, x, x^2, \ldots\}$

Then S is multiplicative and $S \cap I = \emptyset$

By the proposition the maximal among ideals of R that contain I and avoid S is a prime ideal

From $S \cap P = \emptyset$ we get $x \notin P \Rightarrow x \notin \bigcap_{I \subseteq P \text{ prime}} P$

• let say $x \in \sqrt{I}$ and $P \supseteq I$ is prime, then $x^n \in P$
$\overset{P \text{ prime}}{\Rightarrow} x \in P \Rightarrow x \in \bigcap R$

Corollary: If $I \trianglelefteq R$ then $\sqrt{I} \trianglelefteq R$

Proof: By theorem, $\sqrt{I}$ is the intersection of (prime) ideals, so it is an ideal ■

Alternative proof:

$R \cdot \sqrt{I} \subseteq \sqrt{I}$

$x, y \in \sqrt{I}$, say $x^n \in I$ $y^m \in I$

$(x+y)^{m+n-1} = \sum \binom{m+n-1}{k} x^k y^{m+n-1-k}$

if $k \geq n$ then $x^k \in I$

if $k < n$ then $m+n-1-k \geq m \Rightarrow y^{m+n-1-k} \in I$

$\Rightarrow x+y \in \sqrt{I}$

IV Modules

- R ring, R-module $\mathcal{M}$ is $(\mathcal{M},+)$ abelian with scalar multiplication $R\times\mathcal{M}\to\mathcal{M}$

$$(x,m)\mapsto x\cdot m$$

such that a) $x\cdot(m+n)=x\cdot m+xn$ } distributivity

b) $(x+y)\cdot m=xm+ym$

c) $(xy)m=x(ym)$ → asociativity

d) $1\cdot m=m$ ... monicity

A **submodul** is a sub group that is closed under scalar multiplication

$I\trianglelefteq R\Rightarrow I\cdot M$ is submodule of M generated by $x\cdot m\,;\,x\in I,\ m\in M$ modul!!!



- To each $m \in M$ we associate its annihilator
  $\mathrm{Ann}_R(m) := \{x \in R ; x \cdot m = 0\} \lhd R$
- $N \subseteq M \Rightarrow \mathrm{Ann}(M) \subseteq \mathrm{Ann}(N)$ and

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• $\mu_x : \mathcal{M} \longrightarrow \mathcal{M} \qquad x \in R$

$\mu_x : m \longmapsto m \cdot x$

$\mu_R : R \longrightarrow End_{\mathbb{Z}}(\mathcal{M})$

$x \longrightarrow \mu_x$ is a ring homomorphism

$\ker \mu_R = Ann(\mathcal{M})$

conversly if $(N, +)$ is an abelian group

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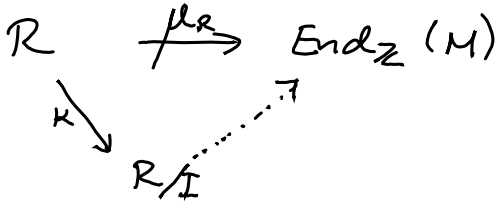
- Algebras: A ring R' is R algebra if $\exists\ R \xrightarrow{\varphi} R'$
  M' R'-module becames R-module by restricting of
  scalars $\quad x \cdot m = \varphi(x) \cdot n$
  $R \quad M \quad R'$

Filip

27.10

$I \lhd R$, $R' = \frac{R}{I}$

• Let M be R-module. When is M R'-module



M is R' module $\Leftrightarrow I \cdot M = 0 \Leftrightarrow I \subseteq \ker \mu_R$

• R' R-algebra with structure map $\varphi : R \to R'$

R'' is subalgebra of R' if R'' is a subring of R' & $\varphi(R) \subseteq R''$

Subalgebra generated by $\{x_\lambda : \lambda \in \Lambda\} \subseteq R'$ is the smallest R-subring containing the elements

Notation : $R[x_\lambda | \lambda \in \Lambda]$

if $|\Lambda| < \infty$ then the subalgebra is finitly generated

Cyclic modules

R-module M is cyclic if $\exists m \in M.\ M = Rm$

Form $\alpha : R \to M$

$x \mapsto xm$ R-module map

α is surjective $\Rightarrow M \cong R/\ker\alpha = R/\mathrm{Ann}(m) = R/\mathrm{Ann}(M)$

- if $I \trianglelefteq R$ then R/I is cyclic R module generated by $1+I$

Cokernel and Coimage

R ring, $\alpha : M \to N$ R-module map

cokernel $\mathrm{coker}\,\alpha := N/\mathrm{im}\,\alpha$

coimage $\mathrm{coim}\,\alpha := M/\ker\alpha$

$\mathrm{coker}\,\alpha = 0 \Leftrightarrow \alpha$ surjective

$\mathrm{coim}\,\alpha \cong \mathrm{im}\,\alpha$

Direct product and direct sums

M_λ $(\lambda \in \Lambda)$ R-modules

- **Direct product** is module $\prod_{\lambda\in\Lambda} M_\lambda = \{(m_\lambda)_{\lambda\in\Lambda} \mid m_\lambda \in M_\lambda\}$
  operations are defined kemponent wise
- **Direct sum** is a submodule of the product
  $\bigoplus_{\lambda\in\Lambda} M_\lambda = \{(m_\lambda);\ m_\lambda \in M_\lambda \text{ and finitly many nonzero}\} \leq \prod_\lambda M_\lambda$

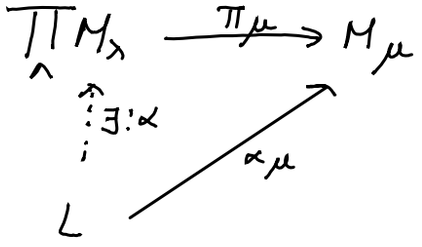
if $|\Lambda| < \infty \Rightarrow \bigoplus M_\lambda = \prod M_\lambda$

if $(\forall \lambda_1, \lambda_2 \in \Lambda \ M_{\lambda_1} = M_{\lambda_2})$ we write $\bigoplus_\Lambda M_\lambda = M^\Lambda$

UL for the product

$\exists \pi_\mu : \prod_\lambda M_\lambda \to M_\mu$

$(m_\lambda)_\lambda \mapsto m_\mu$



$\forall \alpha_\mu : L \to M_\mu \ . \ \exists \alpha : L \to \prod_\Lambda M_\lambda \ . \ \pi_\mu \circ \alpha = \alpha_\mu$

$\alpha(l) = \alpha_\lambda(l)_{\lambda\in\Lambda}$ we write $\alpha = (\alpha_\lambda)_{\lambda\in\Lambda}$

$\Rightarrow \mathrm{Hom}(L, \prod_\Lambda M_\lambda) = \prod_\Lambda \mathrm{Hom}(L, M_\lambda)$

UL for $\oplus$

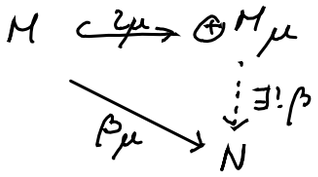
$\exists \iota_\mu : M \longrightarrow \bigoplus_\Lambda M_\lambda$

$m \longmapsto (m_\lambda)_\lambda \quad m_\lambda = 0$ if $\lambda \neq \mu$, m if $\lambda = \mu$

$\forall \beta_\mu : M_\mu \to N$

$\exists ! \beta : \bigoplus M_\mu \to N$

$\beta \circ \iota_\mu = \beta_\mu \quad \forall \mu \in \Lambda$



$\beta((m_\lambda)) = \sum \beta_\lambda(m_\lambda)$

$\beta = \sum \beta_\lambda$

$\Rightarrow \mathrm{Hom}(\bigoplus_\Lambda M_\lambda, N) = \prod_\Lambda \mathrm{Hom}(M_\lambda, N)$

- if $M_\lambda = R$ then $\bigoplus M_\lambda = R^{\wedge}$ is free R-module of $\wedge$
- if $M_\lambda = R$ then $Hom(R^{\wedge}, N) \cong \prod Hom(R, N) \cong \prod_{\wedge} N = N^{\wedge}$
- if M is a module and $S \subseteq M$ then $\langle S \rangle$ is submodule generated by S
- M is finetly generated if there exist finite $S \subseteq M$. $\langle S \rangle = M$

V Exact sequences

$$\dots \; M_i \xrightarrow{\alpha_i} M_{i+1} \xrightarrow{\alpha_{i+1}} M_{i+2} \longrightarrow \dots$$

This sequence is exact at M_i if $\operatorname{im}\alpha_{i-1} = \ker\alpha_i$

if a sequence is exact at every term it is exact

Examples

$0 \longrightarrow L \xrightarrow{\alpha} M \iff \alpha$ injective

$M \xrightarrow{\beta} N \longrightarrow 0 \iff \beta$ surjective

$0 \longrightarrow L \xrightarrow{\alpha} M \xrightarrow{\beta} N \longrightarrow 0$ exact

$0 \longrightarrow L \hookrightarrow L \oplus N \xrightarrow{\pi} N$

Proposition

1) $M_\lambda' \longrightarrow M_\lambda \longrightarrow M_\lambda''$ exact for all λ

* Then $\oplus M_\lambda' \longrightarrow \oplus M_\lambda \longrightarrow \oplus M_\lambda''$ exact and

** $\pi M_\lambda' \longrightarrow \pi M_\lambda \longrightarrow \pi M_\lambda''$

conversly if * or * is exact then $M_\lambda' \to M_\lambda \to M_\lambda''$

2) $0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0$ is exact

short exact sequence. Let $N \subseteq M$

$N' = \alpha^*(N)$ $N'' = \beta_*(N)$

Then $0 \longrightarrow N' \xrightarrow{\alpha|_{N'}} N \xrightarrow{\beta|_N} N''$ is also short exact

Retractions, sections and split exact sequences

- $\rho : M \to N'$ is a retraction of $\alpha : N' \to M$ if $\rho \circ \alpha = 1_{N'}$ (α injective, ρ surjective)
- $\sigma : M'' \to M$ is a section of $\beta : M \to M''$ if $\beta \circ \sigma = 1_{M''}$ (β surj., σ inj.)
- A short exact sequence $N' \xrightarrow{\alpha} M \xrightarrow{\beta} M''$ splits if $\exists \varphi : M \xrightarrow{\cong} N' \oplus M''$ such that

$$\varphi\alpha = \iota_{N'} \quad \text{and} \quad \beta = \pi_{M''} \circ \varphi$$

Proposition: for short exact sequence $0 \to N' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \to 0$:

1) $\Leftrightarrow$ α has a retraction
2) $\Leftrightarrow$ β has a section
3) $\Leftrightarrow$ the sequence splits

Proof Algebra 3

Example: R' R-algebra, M R'-module

$H := \mathrm{Hom}_R(R', M)$

$\alpha : M \longrightarrow H$

$m \longmapsto (x \longmapsto xm)$ R-linear map

$\beta : H \longrightarrow M$

$\varphi \longmapsto \varphi(1)$

β is a retraction of α

This means $0 \longrightarrow M \xrightarrow{\alpha} H \longrightarrow \mathrm{coker}\,\alpha \longrightarrow 0$

splits $H \cong M \oplus \mathrm{coker}\,\alpha$

Snake lemma:

Given a commutative diagram with exact rows

$$\begin{array}{ccccccccc} M' & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & M'' & \longrightarrow & 0 & & \\ \downarrow{\scriptstyle \gamma'} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \gamma''} & & & & \\ 0 & \longrightarrow & N' & \xrightarrow[\alpha']{} & N & \xrightarrow[\beta']{} & N'' & & \end{array}$$

There is a homomorphism $\partial : \ker \gamma'' \longrightarrow \operatorname{coker} \gamma'$

such that $\ker \gamma' \xrightarrow{\bar{\alpha}} \ker \gamma \xrightarrow{\bar{\beta}} \ker \gamma'' \xrightarrow{\partial} \operatorname{coker} \gamma' \xrightarrow{\bar{\alpha}} \operatorname{coker} \gamma \xrightarrow{\bar{\beta}'} \operatorname{coker} \gamma''$ is exact

Moreover: α is injective $\Rightarrow \bar{\alpha}$ is injective

β' surjective $\Rightarrow \bar{\beta}'$ surjective

Proof

$\bar{\alpha} = \alpha|_{\ker \gamma'} : \ker \gamma' \longrightarrow \ker \gamma$

$\bar{\beta} = \beta|_{\ker \gamma} : \ker \gamma \longrightarrow \ker \gamma''$

is codomain right? $m \in \ker \gamma'$

$\gamma(\alpha(m')) = \alpha'(\gamma'(m')) = \alpha(0) = 0$

$m \in \ker \gamma$

$\gamma''(\beta(m)) = \beta'(\gamma(m)) =$

$\bar{\alpha}' : N'/\operatorname{im}\gamma' \longrightarrow M/\operatorname{im}\gamma \qquad [n'] \longrightarrow [\alpha'(n')]$

$\bar{\beta} : M/\operatorname{im}\gamma \longrightarrow N''/\operatorname{im}\gamma'' \qquad [n] \longrightarrow [\beta'(n)]$

is this well defined

if $n_1' - n_2' = \gamma'(m')$

then $\alpha'(n_1') - \alpha'(n_2') = \alpha'(n_1' - n_2') =$

$\alpha'(\gamma(n_1 - n_2)) = \gamma(\alpha(m')) \in \operatorname{im}\gamma$

Step 2) Defining ∂

$m'' \in \ker \gamma''$. since β is surjective,

$\exists\, m \in M.\ \beta(m) = m''$

Then $\beta'(\gamma(m)) = \gamma''(\beta(m)) = \gamma''(m'') = 0$

$\Rightarrow \gamma(m) \in \ker \beta'$

Then $\exists n' \in N'.\ \alpha'(n') = \gamma(m)$

since α is injective n' is unique after picking m

Define $\partial(m'') = [n'] \in N'/\operatorname{im}\gamma'$

Well defined

Suppose m_1, m_2 are lifts of $m'' : \beta(m_j) = m''$

As above there are $\exists n_j'.\ \alpha'(n_j') = \gamma(m_j)$

Claim: $n_1 - n_2 \in \operatorname{im}\gamma'$

Filip

Theorem

1) $M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$ is exact

$\Leftrightarrow 0 \longrightarrow Hom(M'', N) \xrightarrow{\beta^*} Hom(M, N) \xrightarrow{\alpha^*} Hom(M', N)$

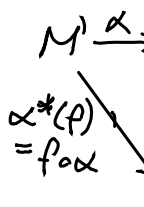
is exact $\forall N$

2) $0 \longrightarrow N' \xrightarrow{\alpha} N \xrightarrow{\beta} N''$ is exact

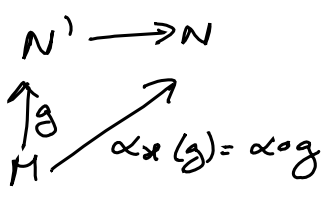
$\Leftrightarrow 0 \longrightarrow Hom(M, N') \xrightarrow{\alpha_*} Hom(M, N) \xrightarrow{\beta_*} Hom(M, N'')$

is exact $\forall M$

what is α^* ?



α_*



Proof:

1) ($\Longrightarrow$) β^* is injective

Recimo da $\beta^*(f) = 0 = f(\beta)$

For any $m'' \in M''$ $\exists m \in M$: $\beta(m) = m''$

Then $f(m'') = f(\beta(m)) = 0 \Rightarrow f = 0$ (surjectivity)

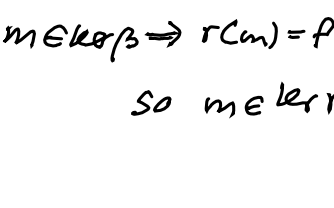
$im\beta^* \subseteq \ker\alpha^*$

$f : M'' \longrightarrow N$

$\alpha^*(\beta^* f) = f \circ \beta \circ \alpha = 0$ ($\beta \circ \alpha = 0$)

$\ker\alpha^* \subseteq im\beta^*$

$g : M \longrightarrow N$. $\alpha^* g = g(\alpha) = 0$



we want f such that $g = f \cdot \beta = \beta^*(f)$

$\beta(m) = m''$

$f(m'') := g(m)$

well defined:

$\beta(m_1) = \beta(m_2) = m''$

exactness then $\beta(m_1 - m_2) = 0$

$\Rightarrow$ $\exists m' \in M'$. $\alpha(m') = m_1 - m_2$

$g(m_1) - g(m_2) = g(m_1 - m_2) = g(\alpha(m')) = 0$

f is R-linear : easy

($\Longleftarrow$) $\beta\alpha = 0$ ($im\alpha \subseteq \ker\beta$)

choose $N = M''$ and apply $\alpha^*\beta^* = 0$ to $id_{M''}$

$0 = \alpha^*(\beta^*(id_{M''})) = \beta \circ \alpha$

β surjective

Test module $N = coker\beta = \frac{M''}{im\beta}$

$q : M'' \longrightarrow N$ quotient map

$q \circ \beta = 0 \Rightarrow \beta^*(q) = 0$

But β^* is injective by exactness

$\Rightarrow q = 0 \Longrightarrow N = 0 \Rightarrow M'' \cong im\beta$

$\ker\beta \subseteq im\alpha$

$M = \frac{M}{im\alpha}$ $\quad r \longrightarrow$ retract map

$r\alpha = 0 \Rightarrow r \in \ker\alpha^* = im\beta^*$

$\Rightarrow \exists f : M'' \longrightarrow N$. $r = \beta^*(f) = f \circ \beta$

if $m \in \ker\beta \Rightarrow r(m) = f(\beta(m)) = 0$

so $m \in \ker r = im\alpha$

$\Rightarrow$

② ($\Rightarrow$) $0 \to N' \xrightarrow{\alpha} N \xrightarrow{\beta} N''$ is exact

$\Rightarrow \alpha$ is injective $\operatorname{im}\alpha = \ker\beta$

Fix M $\quad 0 \to \mathrm{Hom}(M,N') \xrightarrow{\alpha_*} \mathrm{Hom}(M,N) \xrightarrow{\beta_*} \mathrm{Hom}(M,N'')$

is also exact

α_* is injective

$\alpha_*(f) = \alpha \circ f = 0$

$\forall m \in M.\ \alpha(f(m)) = 0 \overset{\alpha \text{ inj}}{\Rightarrow} f(m) = 0 \Rightarrow f = 0$

$\operatorname{im}\alpha_* = \ker\beta_*$

$f : M \to N'$

$\beta_*(\alpha_*)(f) = \beta\alpha(f) = 0 \quad \Rightarrow \operatorname{im}\alpha_* \subseteq \ker\beta_*$

Suppose $g : M \to N$ and $\beta_* g = \beta g = 0$

$$N' \xrightarrow{\alpha} M \xrightarrow{\beta} N''$$

$\exists! f : M \to N'$, with $g : M \to N$

$\forall m \in M.\ g(m) \in \ker\beta = \operatorname{im}\alpha$

$\exists f(m) \in N'.\ g(m) = \alpha(f(m))$

by injectivity of α this element is unique

f R-linear $\quad m_i \in M$ $\quad r, s \in R$

$$\alpha(f(rm_1 + sm_2)) = g(rm_1 + sm_2) =$$

$$= r\,g(m_1) + s\,g(m_2) =$$

$$r\cdot\alpha(f(m_1)) + s\alpha(f(m_2)) = \alpha(rf(m_1) + sf(m_2))$$

α is injective $\Rightarrow$

$\Rightarrow f$ is R-linear Hence $*$ gives $g \to \alpha(f)$ is an

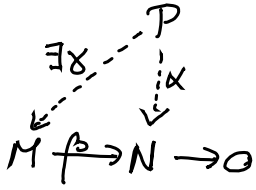
($\Leftarrow$)

Od zdaj naprej pišem samo še izreke, brez dokazov.

V.1. Projective modules

Def: Module P is projective $\Leftrightarrow$

$$\forall \beta \in M \twoheadrightarrow N.\ \forall \alpha : P \to N.\ \exists \gamma : P \to M.\ \alpha = \beta\gamma$$



Examples: Every free module is projective

Exercise: Find a projective module that is not free

Theorem

1) $\Leftrightarrow P$ is projective

2) $\Leftrightarrow \forall$ short exact sequence $0 \to K \to M \to P \to 0$ splits

3) $\Leftrightarrow \exists K.\ K \oplus P$ is free module

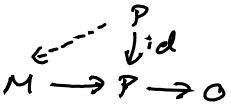
4) $\Leftrightarrow \forall$ exact sequence $N' \to N \to N''$ the sequence $Hom(P,N') \to Hom(P,N) \to Hom(P,N'')$ is exact

5) $\Leftrightarrow \forall \beta : M \twoheadrightarrow N : Hom(P,M) \xrightarrow{\beta_*} Hom(P,N)$

β surj $\Rightarrow$ β_* surj

Proof:

1) $\Rightarrow$ 2)



2) $\Rightarrow$ 3) $\exists \Lambda : R^{\oplus \Lambda} \xrightarrow{\pi} P$ $\quad K = \ker \pi$

$0 \to K \to R^{\oplus \Lambda} \to P \to 0$ splits

$\Rightarrow R^{\oplus \Lambda} = K \oplus P$

3) $\Rightarrow$ 4) $K \oplus P$ is free

$N' \to N \to N''$ is exact

$\Rightarrow \pi N' \to \pi N \to \pi N''$ is exact

$Hom(R^{\oplus \Lambda}, N') \quad Hom(R^{\oplus \Lambda}, N) \quad Hom(R^{\oplus \Lambda}, N'')$

$\Rightarrow Hom(K \oplus P, N') \to Hom(K \oplus P, N') \ldots$ is exact

$\Rightarrow Hom(K,N) \oplus Hom(P,N') \to \ldots$ is exact

$\Rightarrow Hom(P,N') \to \ldots$ is exact

4) $\Rightarrow$ 5) $M \xrightarrow{\beta} N \to 0$ is exact

$\Rightarrow Hom(P,M) \xrightarrow{\beta_*} Hom(P,N) \to 0$ is exact

5) $\Rightarrow$ 1) $\forall \alpha \in Hom(P,N)$ is image under β_*

$\gamma \in Hom(P,M) : \alpha = \beta\gamma$

Cor:

a) $\forall$ R-module M, N, P.

$\mathrm{Hom}(M, \mathrm{Hom}(N, P)) \cong \mathrm{Hom}(M \otimes N, P)$

b) If $M_1 \xrightarrow{\varphi_1} M_2 \xrightarrow{\varphi_2} M_3 \longrightarrow 0$ is exact then

$M \otimes N \xrightarrow{\varphi_1 \otimes id} M_2 \otimes N \xrightarrow{\varphi_2 \otimes id} M_3 \otimes N \longrightarrow 0$ is also exact

($\otimes$ is right exact

V.1 Flat modules

Def: M is flat $\Leftrightarrow$ functor $-\otimes M$ is exact

Eg. since $\otimes$ is right exact, M is flat $\Leftrightarrow$
$\forall N' \hookrightarrow N \quad N' \otimes M \longrightarrow N \otimes M$ is injective

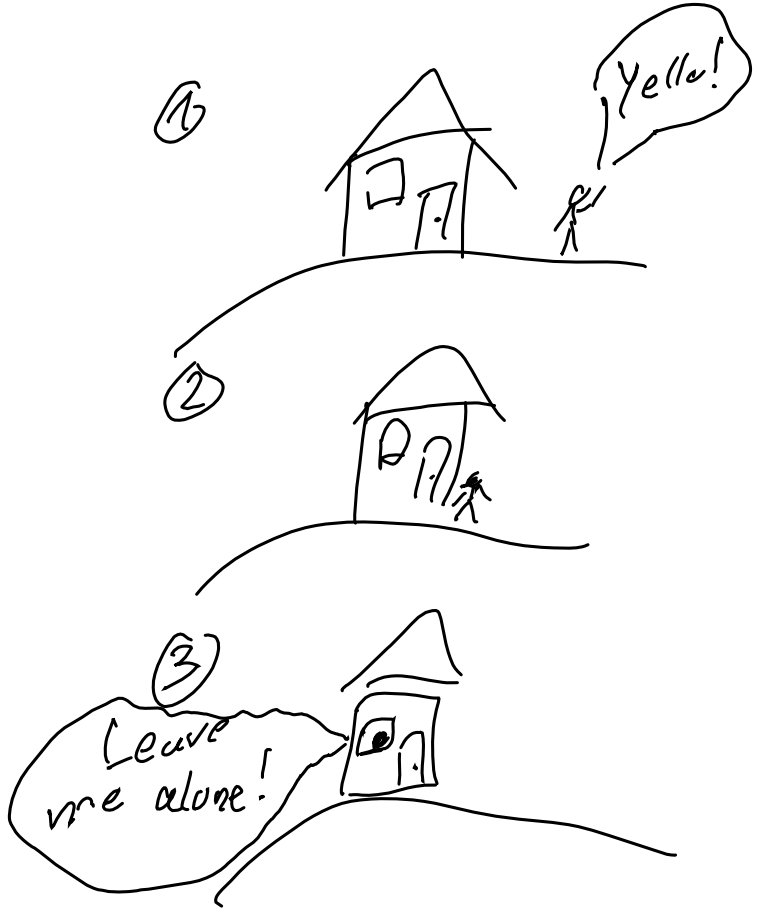
Example: • Free module is flat
• Projective modules are flat

Exercise: Flat module that is not projective

Prop
1) $\Leftrightarrow M$ is flat
2) $\Leftrightarrow \forall u: N' \hookrightarrow N \quad N' \otimes M \xrightarrow{u \otimes id} N \otimes M$ is injective
3) $\Leftrightarrow 0 \longrightarrow N' \longrightarrow N \longrightarrow N''$ is exact $\Rightarrow$

$\Rightarrow 0 \longrightarrow N' \otimes M \longrightarrow N \otimes M \longrightarrow N'' \otimes M \longrightarrow 0$ is exact

Lemma: R-module M is flat $\Leftrightarrow$ every relation of the form $a_1 m_1 + \dots a_n m_n = 0$ $(a_j \in R, m_j \in M)$ can be written in the form $m_i = \sum_{j=1}^{r} b_{ij} n_j$ $(b_{ij} \in R, n_j \in M)$ and $\sum_{i=1}^{n} a_i b_{ij}$ $\forall j \in \{1 \dots r\}$



Theorem:

1) $\Leftrightarrow$ M is flat

2) $\Leftrightarrow$ $\forall$ finitely generated $I \trianglelefteq R$. $I \otimes M \longrightarrow M$ is injective, $a \otimes m \mapsto a \cdot m$

Exercise: prove that $\mathbb{Q}$ is not projective, flat $\mathbb{Z}$-module

V.3 Tensor products of algebras

$f: A \to B$ ring homomorphism

restriction of scalars: B-module becomes A-module

$$a \cdot n := f(a) \cdot n$$
$$a \in A \quad n \in N$$

extension of scalars: M A-module, $M_B = B \otimes_A M$

is an A module and B-module

$b'(b \otimes m) = b'b \otimes m$

R ring, S, T R-algebras with structure hom.

$\varphi: R \to S$, $\psi: R \to T$

$D = S \otimes_R T$ is an R-module

we want $(s \otimes t) \cdot (s' \otimes t') = (ss' \otimes tt')$

Consider the map $\Omega: S \times T \times S \times T \to D$

$(s, t, s', t') \longmapsto ss' \otimes tt'$

Ω is R-linear in each factor so it induces a map

$D \otimes D = S \otimes T \otimes S \otimes T \longrightarrow D$

$s \otimes t \otimes s' \otimes t' \longmapsto ss' \otimes tt'$

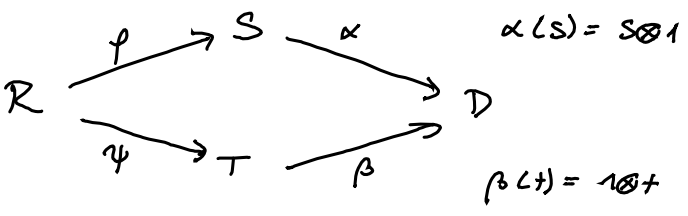
$\Rightarrow$ D is a commutative ring w.r.t identity $1 \otimes 1$

it is also an R algebra via

$R \longrightarrow D$

$r \mapsto \varphi(r) \otimes \psi(r)$

D = tensor product of S and T



This diagram commutes

Example: • $R[x] \otimes R[y] \cong R[x,y]$

$$\Phi : R[x] \otimes R[y] \longrightarrow R[x,y]$$
$$f \otimes g \longmapsto f \cdot g$$

$$\Psi : R[x,y] \longrightarrow R[x] \otimes R[y]$$
$$\sum a_{ij} x^i y^j \longmapsto \sum a_{ij} x^i \otimes y^j$$

check $\Psi \circ \Phi$ and $\Phi \circ \Psi$

• $\mathbb{Q}(i) \otimes_{\mathbb{Q}} \mathbb{Q}(i)$

$(1 \otimes 1 - i \otimes i)(1 \otimes 1 + i \otimes i) = 0$

two fields, tenser is not a field

Exercise: What would be $F(x) \otimes_F F(y)$ vs $F(x,y)$